

## **Lecture 5**

# **Database Logical Design**

# Database Logical Design

- Logical database design is the process of **transforming** the **conceptual data model** into a **logical data model**.

# The Relational Data Model

- The relational data model was first introduced in 1970. Commercial products of **Data Base Management Systems** (DBMS) based on the relational data models (RDBMS) started to appear about 1980 from numerous vendors.
- Today RDBMSs have become the **dominant technology** for data management, and there are literally hundreds of RDBMS products for computers ranging from personal computers to mainframes.
- The relational **data model represents data in the form of tables** (called **relations**). Since an entity (in the conceptual data model) is a person, place, object, event, or concept about which we wish to maintain data, we can understand that entities in the conceptual data model will be mapped to (or replaced by) tables.

# The Relational Data Model (cont.)

- The relational data model consists of the following three components:
  - *Data structure*: Data are organized in the form of **two-dimensional tables** with rows and columns (called relations).
  - *Data manipulation*: Powerful operations (using the **SQL language** or processing programs) are used to manipulate data stored in the relations.
  - *Data integrity*: Facilities are included to specify **business rules** that maintain the integrity of data when they are manipulated.

# Relational Data Structure

- A relation is a **named, two-dimensional table of data**.
- Each **relation** (or table) consists of a set of **named columns** and an arbitrary number of **unnamed rows**.
- An **attribute**, is a **named column** of a table (or relation).
- Each **row** of a table corresponds **to a record** that contains data (attribute) values for a single entity. We must be able to **store and retrieve a row** of data in a relation, based on the **data values** stored in that row.
- To achieve this goal, **every relation must have a primary key**.
- A **primary key** is an attribute (or combination of attributes) **that uniquely identifies each row** in a relation. We designate a primary key by underlining the attribute name.

# Primary Key

- Each table must **have a primary key**.
- Each table must have **only one primary key**.
- PK means primary key.
- PK values must **never be null** (NN).
- PK values must **never be duplicate** (ND).

# Primary Key

- PK values must **never be changed**.
- PKs do not influence **column order**.
- PKs do not influence **row order**.
- PKs do not influence **row access**.
- PKs may be **user assigned** (UA) or **system assigned** (SA)

# Foreign Key

- A foreign key is a column or a group of columns that is **a primary key elsewhere**.
- FK means foreign key.
- FK values **must refer to existing PK values**.

| Department |           |
|------------|-----------|
| Dept Code  | Dept Name |
| PK, UA     | NN, ND    |
| SA         | Sales     |
| PA         | Payroll   |
| RE         | Research  |
| MK         | Marketing |
| GA         | Graphics  |

| Employee        |           |               |               |
|-----------------|-----------|---------------|---------------|
| Employee Number | Dept Code | Employee Name | Salary Amount |
| PK, SA          | FK        | NN            |               |
| 1               | SA        | Drucker       | 12,000        |
| 2               | PA        | Fonebone      |               |
| 3               |           | Gaines        | 36,000        |
| 4               |           | Newman        | 24,000        |
| 5               | PA        | Jane          | 24,000        |

# Properties of Relations (or Tables)

- We have defined relations as two-dimensional tables of data.
- However, **not all tables are relations**.
- **Relations have several properties** that distinguish them from non-relational tables:
  - **Each relation** (or table) in a database **has a unique name**.
  - **An entry** at the intersection of each row and column **is atomic** (or single-valued). There can be no multi-valued attributes in a relation. This is called **atomicity**.

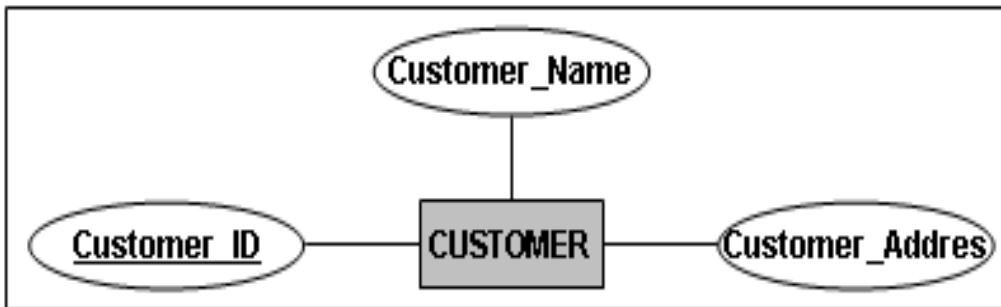
# Properties of Relations (or Tables)

- **Each row is unique**; no two rows in a relation are identical.
- **Each attribute** (or column) within a table **has a unique name**.
- **The sequence of columns** (left to right) **is insignificant**. The columns of a relation can be interchanged without changing the meaning or use of the relation.
- **The sequence of rows** (top to bottom) **is insignificant**. As with columns, the rows of a relation may be interchanged or stored in any sequence.

# Transforming E-R Diagrams Into Relations (or Tables)

- During logical design you **transform the E-R diagrams** that were developed during conceptual design **into relational database schemas**.
- The **inputs** to this process are the entity-relationship diagrams.
- The **outputs** are the relational schemas.
- Transforming (or mapping) E-R diagrams to relations is a **relatively straight forward process** with a well-defined set of rules.
- In fact, many **CASE tools** can automatically perform many of the conversion steps. However, it is important to understand the steps of this process.

# Mapping Regular Entity

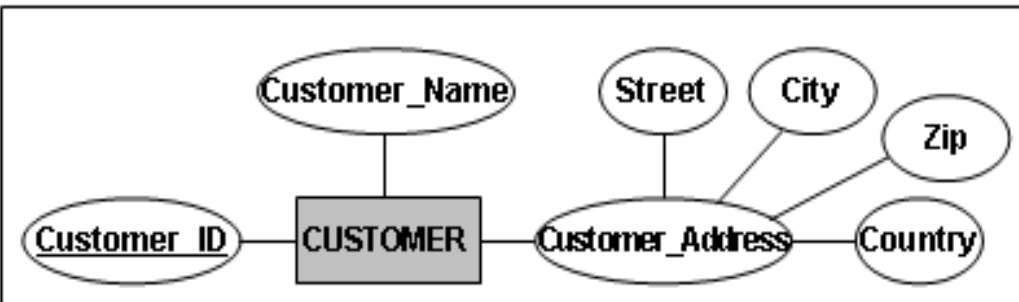


CUSTOMER

| <u>Customer_ID</u> | Customer_Name | Customer_Address |
|--------------------|---------------|------------------|
|--------------------|---------------|------------------|

| Customer_ID | Customer_name        | Customer_Address                          |
|-------------|----------------------|-------------------------------------------|
| 1           | Contemporary Casuals | 1355 S. Hines Blvd, London U.K.           |
| 2           | Value Furniture      | 151 Allard Ave, Paris, France.            |
| 3           | Home Furnishings     | 1900 Kiowai St., Berlin, Germany.         |
| 4           | Eastern Furniture    | 192 Ramsis St., Cairo, Egypt.             |
| 5           | Impressions          | 55 Khaled Ebn Alwalid, Alexandria, Egypt. |
| 6           | Furniture Gallery    | 325 Flatiron St., Rome, Italy.            |

# Mapping Composite Attribute



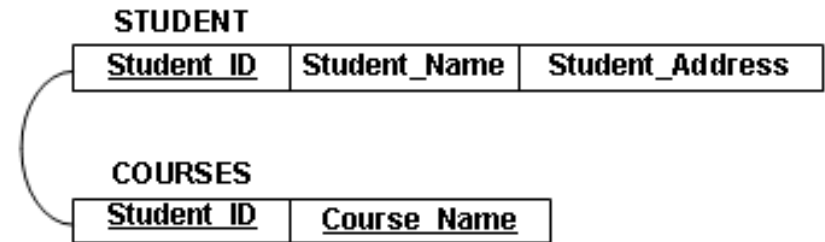
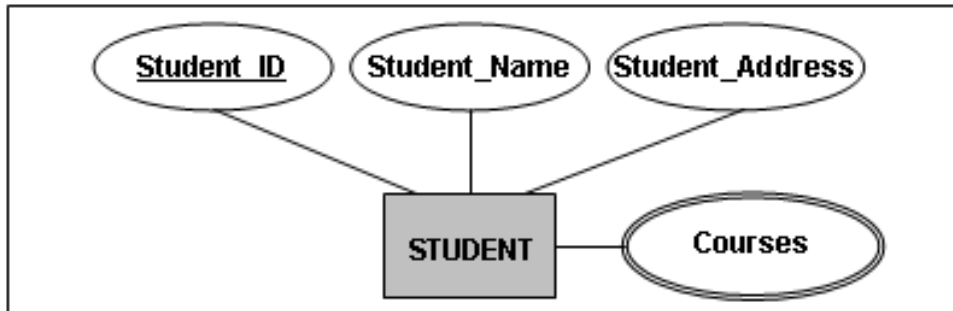
**CUSTOMER**

| <u>Customer_ID</u> | Customer_Name | Street | City | Zip | Country |
|--------------------|---------------|--------|------|-----|---------|
|--------------------|---------------|--------|------|-----|---------|

| Customer_ID | Customer_name        | Street                 | City       | Country | Zip   |
|-------------|----------------------|------------------------|------------|---------|-------|
| 1           | Contemporary Casuals | 1355 S. Hines Blvd.    | London     | U.K.    | 32601 |
| 2           | Value Furniture      | 151 Allard Ave         | Paris      | France  | 75094 |
| 3           | Home Furnishings     | 1900 Kiowai St.        | Berlin     | Germany | 12209 |
| 4           | Eastern Furniture    | 192 Ramsis St.         | Cairo      | Egypt   | 27008 |
| 5           | Impressions          | 55 Khaled Ebn Alwalid. | Alexandria | Egypt   | 94206 |
| 6           | Furniture Gallery    | 325 Flatiron St.       | Rome       | Italy   | 80514 |

# Mapping Multi-valued Attribute for an Entity

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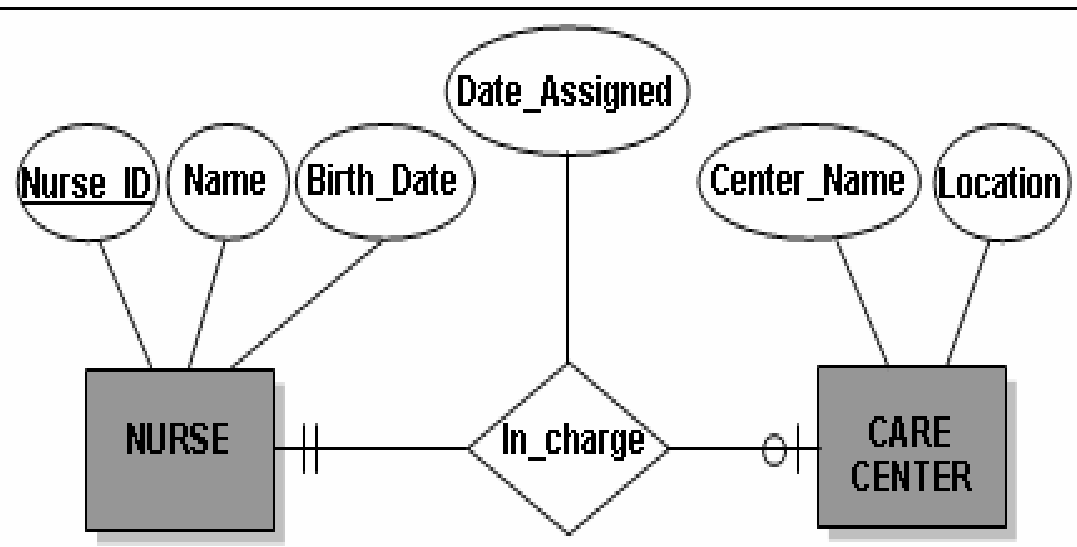
| Student_ID | Student_Name    | Student_Address |                |
|------------|-----------------|-----------------|----------------|
| 1          | Iman Mostafa    |                 | SQL            |
| 2          | Ashraf Zaky     |                 | 1 DataBase     |
| 3          | Ibrahim Saad    |                 | 2 Visual Basic |
| 4          | Moataz Ahmad    |                 | 2 JAVA         |
| 5          | Shereen Mahmoud |                 | 2 DataBase     |
| 6          | Nermeen Hesham  |                 | 3 MIS          |
| 7          | Jassmin Mostafa |                 |                |
| 8          | Rasha Omar      |                 | 8 DataBase     |
| 9          | Mostafa Younes  |                 | 9 DataBase     |
| 10         | Mahmoud Farag   |                 | 10 DataBase    |
| 11         | Jassmin Adel    |                 | 11 DataBase    |
| 12         | Hesham Sief     |                 | 12 DataBase    |
| 13         | Maha Mostafa    |                 | 13 DataBase    |
| 14         | Fadi Talaat     |                 | 14 DataBase    |
| 15         | Dalia Omar      |                 | 15 DataBase    |
| 16         | Mahmoud Essam   |                 | 16 DataBase    |
| 17         | Alaa Hatem      |                 | 17 DataBase    |
| 18         | Menna Younes    |                 | 18 DataBase    |

- Not the best senario

- Use M:N Relation

# Mapping Binary One-to-One Relationship

5



NURSE

| <u>Nurse_ID</u> | Name | Birth_Date |
|-----------------|------|------------|
|-----------------|------|------------|

CARE CENTER

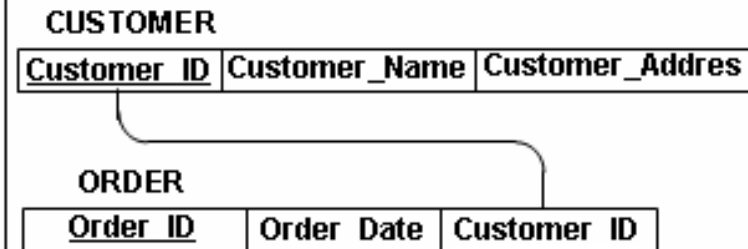
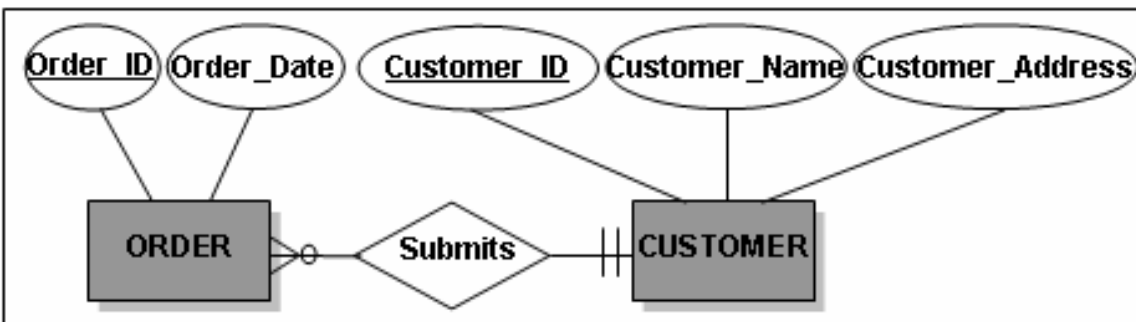
| <u>Center_Name</u> | Location | Nurse_ID | Date_Assigned |
|--------------------|----------|----------|---------------|
|--------------------|----------|----------|---------------|

| Nurse_ID | Name          | Birth_Date |
|----------|---------------|------------|
| 1        | Nadia Kamel   | 01/01/1980 |
| 2        | Fatma Samir   | 02/05/1979 |
| 3        | Samira Ezzat  | 22/03/1985 |
| 4        | Nermin Fadel  | 13/07/1978 |
| 5        | Samia Hassan  | 20/06/1981 |
| 6        | Karima Aly    | 28/02/1980 |
| 7        | Laila Mostafa | 31/12/1981 |

| Center_Name           | Location     | Nurse_ID | Date_Assigned |
|-----------------------|--------------|----------|---------------|
| Pediatric Center      | First Floor  | 1        | 01/01/2002    |
| Surgical Center 1     | Second Floor | 4        | 02/01/2002    |
| Intensive Care Center | Third Floor  | 5        | 03/02/2002    |
| Reception Center      | Ground Floor | 7        | 05/03/2002    |
| Surgical Center 2     | Second Floor | 2        | 02/01/2002    |

# Mapping Binary One-to-Many Relationship

5



| Order_ID | Order_Date | Customer_ID |
|----------|------------|-------------|
| 1001     | 21/10/1998 | ← 1         |
| 1002     | 21/10/1998 | ← 8         |
| 1003     | 22/10/1998 | ← 15        |
| 1004     | 22/10/1998 | ← 5         |
| 1005     | 24/10/1998 | ← 3         |
| 1006     | 27/10/1998 | ← 2         |
| 1007     | 27/10/1998 | ← 11        |
| 1008     | 30/10/1998 | ← 12        |
| 1009     | 05/11/1998 | ← 4         |
| 1010     | 05/11/1998 | ← 1         |

| Customer_ID | Customer_name            | Customer_address      |
|-------------|--------------------------|-----------------------|
| 1           | Contemporary Casuals     | 1355 S. Hines Blvd.   |
| 2           | Value Furniture          | 15145 S.W. 17th St.   |
| 3           | Home Furnishings         | 1900 Allard Ave.      |
| 4           | Eastern Furniture        | 1925 Beltline Rd.     |
| 5           | Impressions              | 5585 Westcott Ct.     |
| 6           | Furniture Gallery        | 325 Flatiron Dr.      |
| 7           | Period Furnishings       | 394 Rainbow Dr.       |
| 8           | California Classics      | 816 Peach Rd.         |
| 9           | M & H Casual Furniture   | 3709 First Street     |
| 10          | Seminole Interiors       | 2400 Rocky Point Dr.  |
| 11          | American Euro Lifestyles | 2424 Missouri Ave. N. |
| 12          | Battle Creek Furniture   | 345 Capitol Ave. SW   |
| 13          | Heritage Furnishings     | 66789 College Ave.    |
| 14          | Kaneohe Homes            | 112 Kiowai St.        |
| 15          | Mountain Scenes          | 4132 Main Street      |

# Mapping Binary Many-to-Many Relationship

5

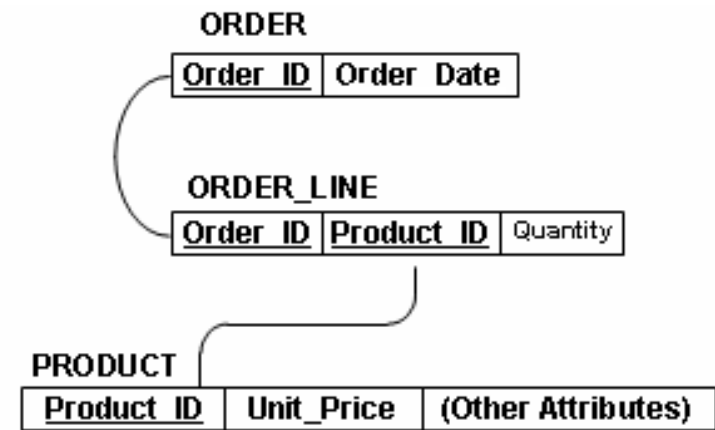
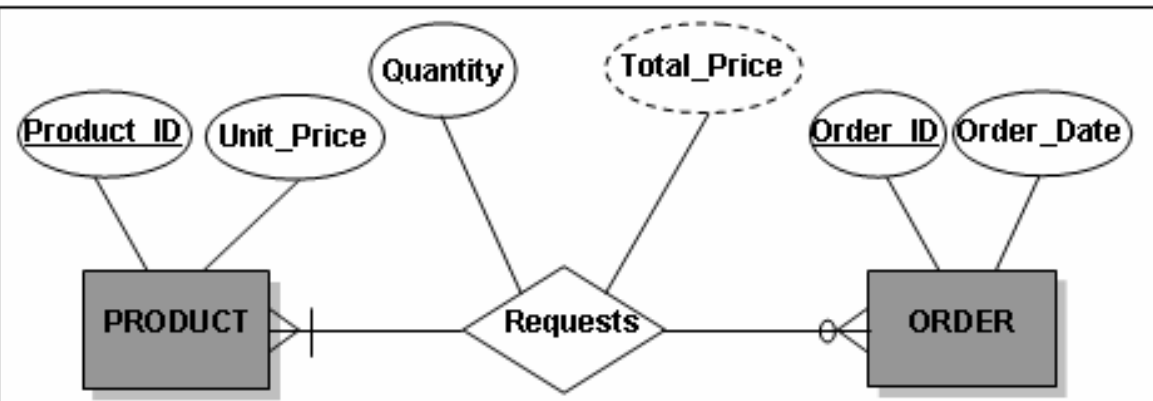


Table PRODUCT

| Product_ID | Product_Name         | Unit_Price | Prod  |
|------------|----------------------|------------|-------|
| 1          | End Table            | \$175.00   | Cherr |
| 2          | Coffee Table         | \$200.00   | Natur |
| 3          | Computer Desk        | \$375.00   | Natur |
| 4          | Entertainment Center | \$650.00   | Natur |
| 5          | Writer's Desk        | \$325.00   | Cherr |
| 6          | 8-Drawer Desk        | \$750.00   | White |
| 7          | Dining Table         | \$800.00   | Natur |
| 8          | Computer Desk        | \$299.00   | Walni |

| Product_ID | Order_ID | Quantity |
|------------|----------|----------|
| 1          | 1001     | 2        |
| 2          | 1001     | 2        |
| 4          | 1001     | 1        |
| 3          | 1002     | 6        |
| 3          | 1003     | 3        |
| 6          | 1004     | 2        |
| 8          | 1004     | 2        |
| 4          | 1005     | 4        |
| 4          | 1006     | 1        |
| 5          | 1006     | 2        |
| 7          | 1006     | 2        |
| 1          | 1007     | 3        |
| 2          | 1007     | 2        |
| 3          | 1008     | 3        |
| 8          | 1008     | 3        |
| 4          | 1009     | 2        |
| 7          | 1009     | 3        |
| 8          | 1010     | 10       |

Table ORDER

| Order_ID | Order Date | Customer_ID |
|----------|------------|-------------|
| 1001     | 21/10/1998 | 1           |
| 1002     | 21/10/1998 | 8           |
| 1003     | 22/10/1998 | 15          |
| 1004     | 22/10/1998 | 5           |
| 1005     | 24/10/1998 | 3           |
| 1006     | 27/10/1998 | 2           |
| 1007     | 27/10/1998 | 11          |
| 1008     | 30/10/1998 | 12          |
| 1009     | 05/11/1998 | 4           |
| 1010     | 05/11/1998 | 1           |

# Mapping Associative Entities (Identifier Not Assigned)

5

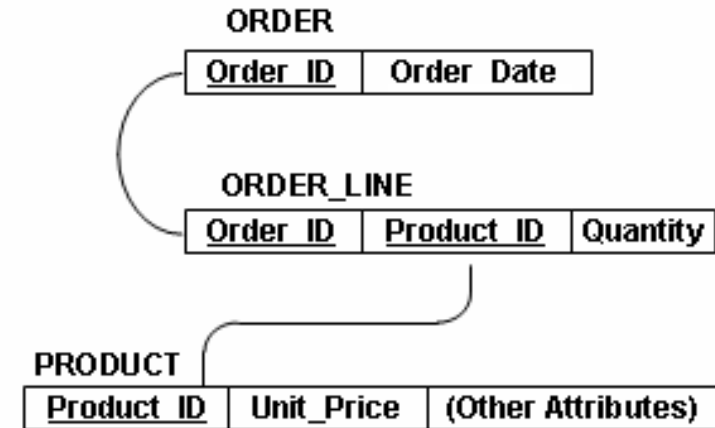
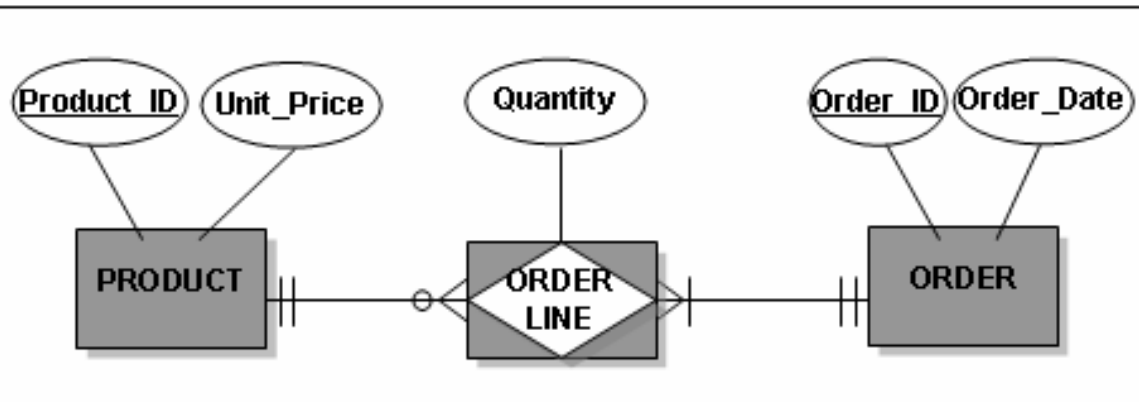


Table PRODUCT

| Product_ID | Product_Name         | Unit_Price | Prod  |
|------------|----------------------|------------|-------|
| 1          | End Table            | \$175.00   | Cherr |
| 2          | Coffee Table         | \$200.00   | Natur |
| 3          | Computer Desk        | \$375.00   | Natur |
| 4          | Entertainment Center | \$650.00   | Natur |
| 5          | Writer's Desk        | \$325.00   | Cherr |
| 6          | 8-Drawer Desk        | \$750.00   | White |
| 7          | Dining Table         | \$800.00   | Natur |
| 8          | Computer Desk        | \$250.00   | Walni |

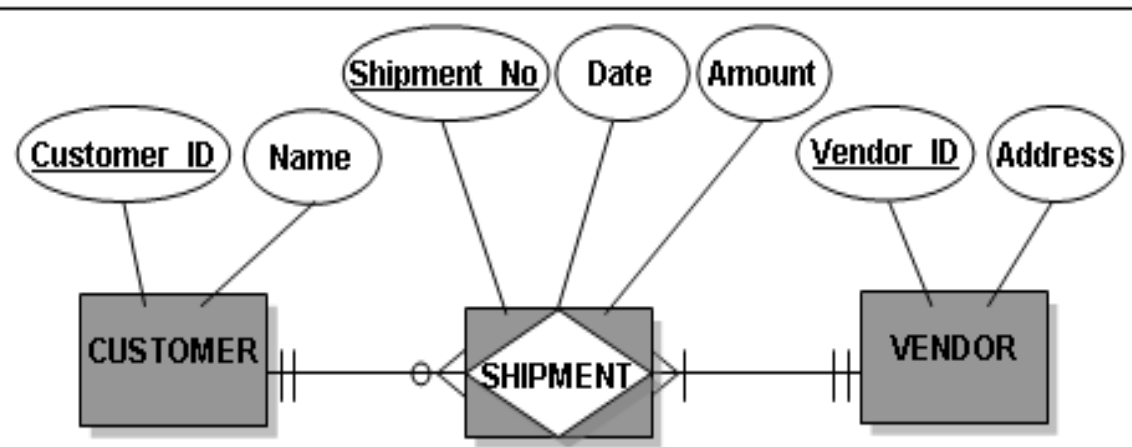
Table ORDER\_LINE

| Product_ID | Order_ID | Quantity |
|------------|----------|----------|
| 1          | 1001     | 2        |
| 2          | 1001     | 2        |
| 4          | 1001     | 1        |
| 3          | 1002     | 5        |
| 3          | 1003     | 3        |
| 6          | 1004     | 2        |
| 8          | 1004     | 2        |
| 4          | 1005     | 4        |
| 4          | 1006     | 1        |
| 5          | 1006     | 2        |
| 7          | 1006     | 2        |
| 1          | 1007     | 3        |
| 2          | 1007     | 2        |
| 3          | 1008     | 3        |
| 8          | 1008     | 3        |
| 4          | 1009     | 2        |
| 7          | 1009     | 3        |
| 8          | 1010     | 10       |

Table ORDER

| Order_ID | Order_Date | Customer_ID |
|----------|------------|-------------|
| 1001     | 21/10/1998 | 1           |
| 1002     | 21/10/1998 | 8           |
| 1003     | 22/10/1998 | 15          |
| 1004     | 22/10/1998 | 5           |
| 1005     | 24/10/1998 | 3           |
| 1006     | 27/10/1998 | 2           |
| 1007     | 27/10/1998 | 11          |
| 1008     | 30/10/1998 | 12          |
| 1009     | 05/11/1998 | 4           |
| 1010     | 05/11/1998 | 1           |

# Mapping Associative Entities (Identifier is Assigned)



## CUSTOMER

| <u>Customer ID</u> | Name | (Other Attributes) |
|--------------------|------|--------------------|
|--------------------|------|--------------------|

## SHIPMENT

| <u>Shipment ID</u> | Customer ID | Vendor ID | Date | Amount |
|--------------------|-------------|-----------|------|--------|
|--------------------|-------------|-----------|------|--------|

## VENDOR

| <u>Vendor ID</u> | Address | (Other Attributes) |
|------------------|---------|--------------------|
|------------------|---------|--------------------|

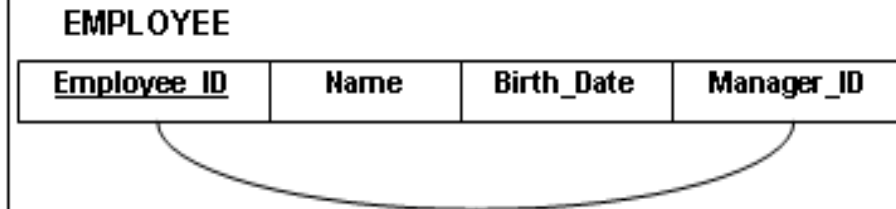
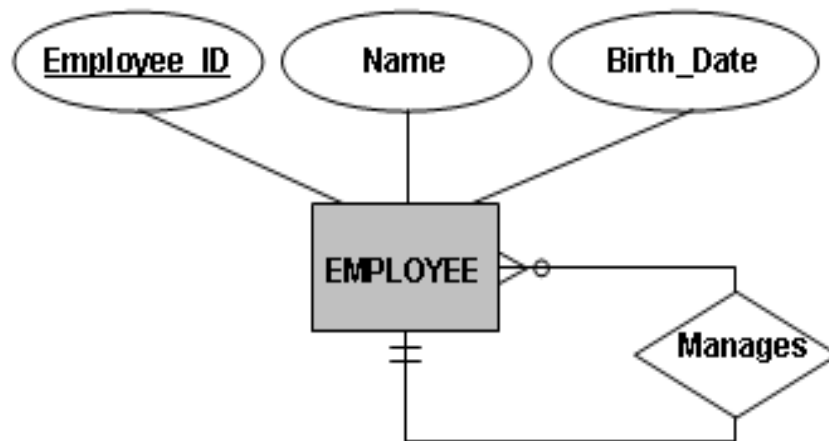
| Customer ID | Customer name        | Customer address    |
|-------------|----------------------|---------------------|
| 1           | Contemporary Casuals | 1355 S. Hines Blvd. |
| 2           | Value Furniture      | 15145 S.W. 17th St. |
| 3           | Home Furnishings     | 1900 Allard Ave.    |
| 4           | Eastern Furniture    | 1925 Beltline Rd.   |

| Vendor ID | Vendor address     | Vendor name             |
|-----------|--------------------|-------------------------|
| 1         | 7708 W Broadway    | L & M Hardwoods         |
| 2         | 5545 Rolling Hills | Building Blocks         |
| 3         | 136 Pearl Avenue   | McKnight Hardware       |
| 4         | 5672 E. Druid Road | Seattle Forest Products |
| 5         | 718 Moss Creek Dr  | Axley Lumber Co.        |
| 6         | 2620 36th St. N    | McEwen Lumber Co.       |

| Shipment No | Customer ID | Vendor ID | Date       | Amount |
|-------------|-------------|-----------|------------|--------|
| 12022       | 1           | 4         | 01/07/2001 | 3      |
| 12023       | 2           | 3         | 02/01/2002 | 2      |
| 12024       | 1           | 3         | 01/03/2001 | 15     |
| 12025       | 3           | 5         | 20/03/2001 | 24     |
| 12026       | 2           | 2         | 15/07/2001 | 8      |
| 12027       | 4           | 1         | 26/11/2000 | 0      |

# Mapping Unary One-to-Many Relationship

5

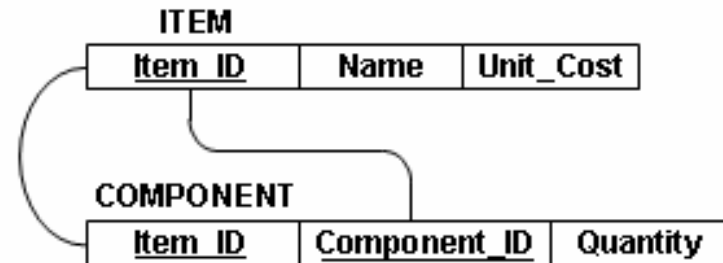
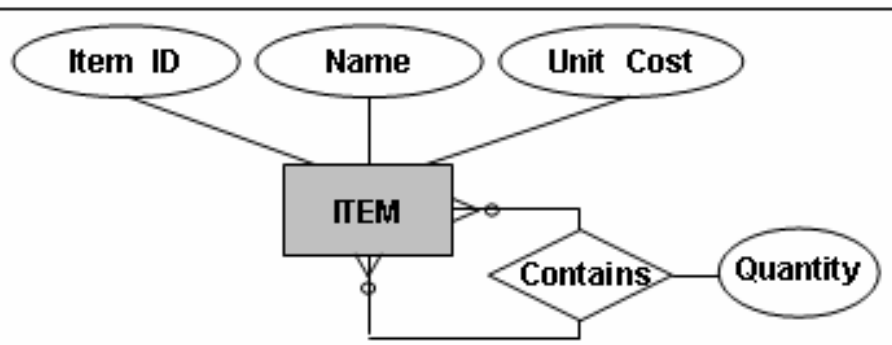


| Employee_ID | Name          | Birth_Date | Manager_ID |
|-------------|---------------|------------|------------|
| 8           | Kamel Reda    | 14/10/1966 | 8          |
| 5           | Mohammad Adel | 28/11/1970 | 8          |
| 1           | Magdy Foad    | 02/05/1975 | 5          |
| 3           | Saber Fathy   | 06/11/1976 | 8          |
| 2           | Naser Samir   | 04/08/1977 | 5          |
| 6           | Wagdy Sabry   | 15/04/1979 | 8          |
| 4           | Anwar Mahmoud | 19/03/1980 | 5          |
| 7           | Hany Galal    | 11/06/1981 | 5          |
| 9           | Tamer Hassan  | 27/12/1982 | 1          |

| Employee_ID | Name          | Birth_Date | Manager_ID |
|-------------|---------------|------------|------------|
| 8           | Kamel Reda    | 14/10/1966 | 8          |
| 5           | Mohammad Adel | 28/11/1970 | 8          |
| 1           | Magdy Foad    | 02/05/1975 | 5          |
| 3           | Saber Fathy   | 06/11/1976 | 8          |
| 2           | Naser Samir   | 04/08/1977 | 5          |
| 6           | Wagdy Sabry   | 15/04/1979 | 8          |
| 4           | Anwar Mahmoud | 19/03/1980 | 5          |
| 7           | Hany Galal    | 11/06/1981 | 5          |
| 9           | Tamer Hassan  | 27/12/1982 | 1          |

# Mapping Unary Many-to-Many Relationship

5



| Item_ID | Name            | Unit_Cost   |
|---------|-----------------|-------------|
| 1       | Computer ST1200 |             |
| 2       | Hard Disk 40G   | 450.00 LE   |
| 3       | Floppy Disk     | 55.00 LE    |
| 4       | CD ROM Drive    | 150.00 LE   |
| 5       | CPU 1.2GHz      | 750.00 LE   |
| 6       | Memory 128M     | 90.00 LE    |
| 7       | Modem 56K       | 70.00 LE    |
| 8       | Sound Card      | 70.00 LE    |
| 9       | Laser Printer   | 1,400.00 LE |
| 10      | Scanner         | 400.00 LE   |
| 11      | VGA Card        | 180.00 LE   |
| 12      | Network Card    | 60.00 LE    |
| 13      | Keyboard        | 25.00 LE    |
| 14      | Mouse           | 40.00 LE    |
| 15      | Mother Board    | 370.00 LE   |
| 16      | AT Case         | 90.00 LE    |
| 17      | Computer ST1400 |             |
| 18      | CPU 1.4GHz      | 900.00 LE   |
| 19      | TV Tuner Card   | 185.00 LE   |
| 20      | CD R/W Drive    | 650.00 LE   |
| 21      | Plotter         | 1,500.00 LE |

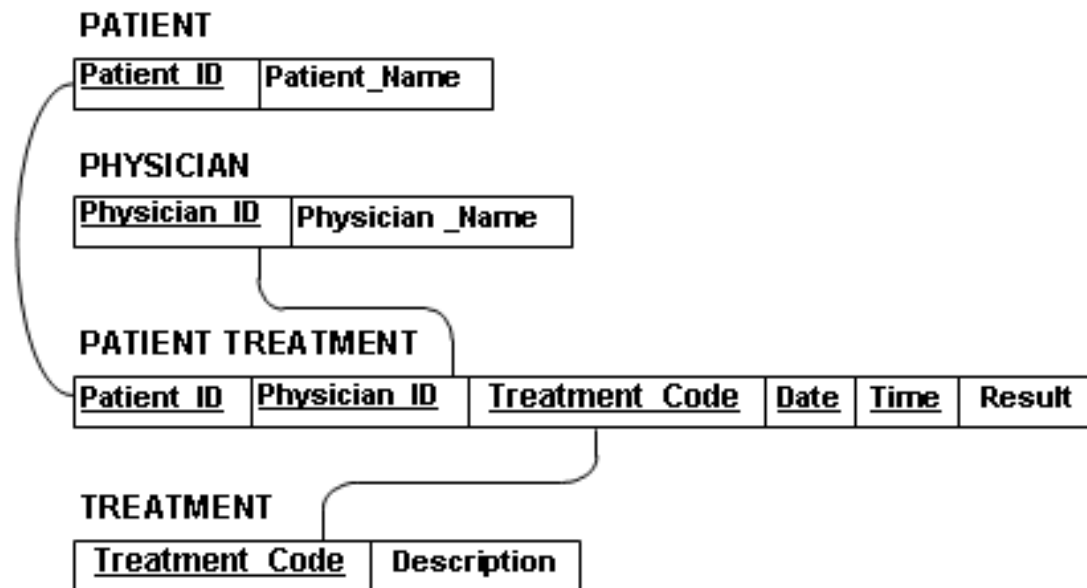
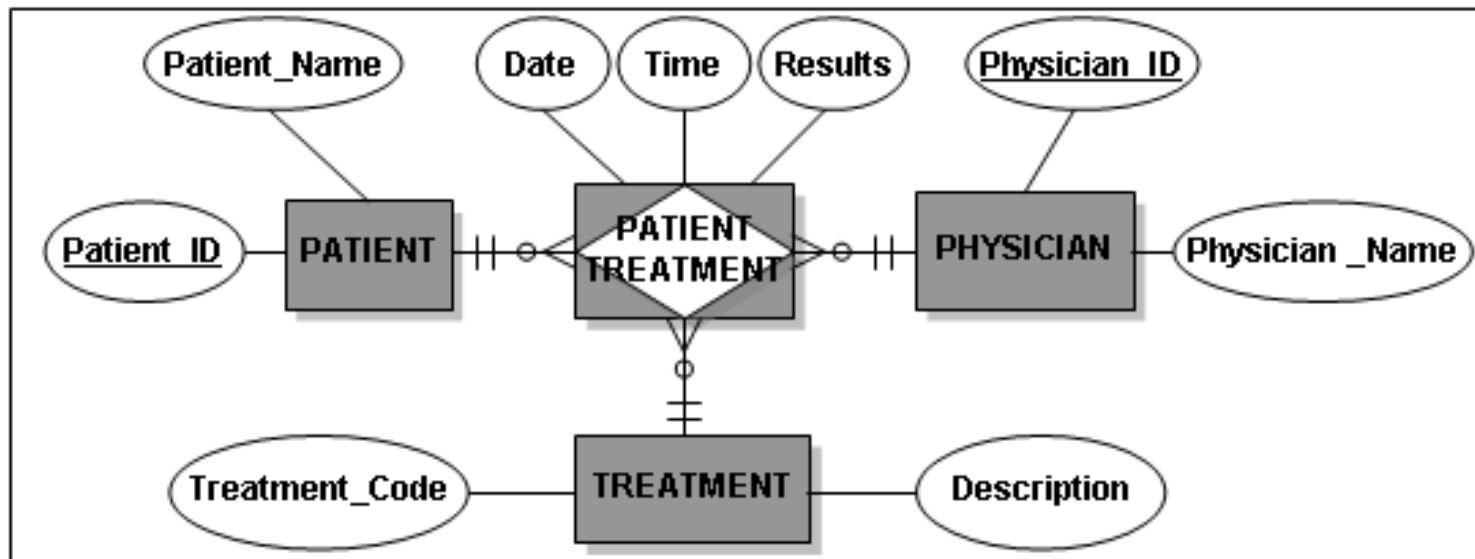
| Item_ID | Component_ID | Quantity |
|---------|--------------|----------|
| 1       | 2            | 2        |
| 1       | 3            | 1        |
| 1       | 4            | 1        |
| 1       | 5            | 1        |
| 1       | 6            | 2        |
| 1       | 7            | 1        |
| 1       | 11           | 1        |
| 1       | 13           | 1        |
| 1       | 14           | 1        |
| 1       | 15           | 1        |
| 1       | 16           | 1        |
| 17      | 2            | 3        |
| 17      | 3            | 1        |
| 17      | 6            | 3        |
| 17      | 9            | 1        |
| 17      | 11           | 1        |
| 17      | 13           | 1        |
| 17      | 14           | 1        |
| 17      | 15           | 1        |
| 17      | 16           | 1        |
| 17      | 18           | 1        |
| 21      | 20           | 2        |

| Item_ID | Name            | Unit_Cost   |
|---------|-----------------|-------------|
| 1       | Computer ST1200 |             |
| 2       | Hard Disk 40G   | 450.00 LE   |
| 3       | Floppy Disk     | 55.00 LE    |
| 4       | CD ROM Drive    | 150.00 LE   |
| 5       | CPU 1.2GHz      | 750.00 LE   |
| 6       | Memory 128M     | 90.00 LE    |
| 7       | Modem 56K       | 70.00 LE    |
| 8       | Sound Card      | 70.00 LE    |
| 9       | Laser Printer   | 1,400.00 LE |
| 10      | Scanner         | 400.00 LE   |
| 11      | VGA Card        | 180.00 LE   |
| 12      | Network Card    | 60.00 LE    |
| 13      | Keyboard        | 25.00 LE    |
| 14      | Mouse           | 40.00 LE    |
| 15      | Mother Board    | 370.00 LE   |
| 16      | AT Case         | 90.00 LE    |
| 17      | Computer ST1400 |             |
| 18      | CPU 1.4GHz      | 900.00 LE   |
| 19      | TV Tuner Card   | 185.00 LE   |
| 20      | CD R/W Drive    | 650.00 LE   |
| 21      | Plotter         | 1,500.00 LE |

# Mapping Ternary (and n-ary) Relationship

5



# Mapping Ternary (and n-ary) Relationship (Continue)

5

| Patient_ID | Patient_Name |
|------------|--------------|
| 1          | Samia Hosny  |
| 2          | Kamel Ezzat  |
| 3          | Magdy Foad   |
| 4          | Wagdy Sabry  |
| 5          | Nadia Samir  |
| 6          | Nehal Omar   |
| 7          | Walid Hassan |
| 8          | Karim Hosny  |
| 9          | Hamed Aly    |

| Patient_ID | Physician_ID | Treatment_Code | Date       | Time        | Result        |
|------------|--------------|----------------|------------|-------------|---------------|
| 1          | 1            | 3              | 01/01/2002 | 12:05:00 PM | Good          |
| 2          | 5            | 5              | 02/02/2002 | 09:15:00 AM | OK            |
| 3          | 6            | 4              | 02/03/2002 | 04:30:00 PM | Not known     |
| 4          | 1            | 1              | 11/01/2002 | 11:30:00 AM | Very Good     |
| 5          | 6            | 2              | 12/03/2002 | 11:00:00 AM | Excellent     |
| 5          | 7            | 5              | 20/01/2002 | 08:00:00 PM | Nice Progress |
| 6          | 2            | 1              | 10/03/2002 | 05:00:00 PM | Good          |
| 7          | 8            | 1              | 03/02/2002 | 05:22:00 PM | Not Enough    |
| 8          | 8            | 5              | 02/04/2002 | 10:00:00 AM | Normal        |
| 8          | 8            | 6              | 05/02/2001 | 06:00:00 PM | OK            |

| Physician_ID | Physician_Name |
|--------------|----------------|
| 1            | Samir Galal    |
| 2            | Saber Fathy    |
| 3            | Kamel Reda     |
| 4            | Wasfy Azeez    |
| 5            | Ashraf Waly    |
| 6            | Ayman Lotfy    |
| 7            | Omayma Salem   |
| 8            | Omar Fahmy     |
| 9            | Amin Fikry     |

| Treatment_Code | Description            |
|----------------|------------------------|
| 1              | Cidophage 2t per day   |
| 2              | Senalax 1t per day     |
| 3              | Aspirin 3t per day     |
| 4              | Vitamin B12 3t per day |
| 5              | Complete Bold analysis |
| 6              | Ampicilin 1t each 6hr  |
| 8              | Cut Scan               |
| 9              | Electro Cardiograph    |

# Data Modeling Concepts: Generalization

- **Generalization** – a concept wherein the attributes that are **common to several types** of an entity are **grouped into** their own entity.
- **Supertype** – an entity whose instances store attributes that are **common** to one or more entity subtypes.
- **Subtype** – an entity whose instances **may inherit common attributes** from its entity supertype
  - And then **add other attributes** unique to the subtype.

# Data Modeling Concepts: Subsetting Criteria

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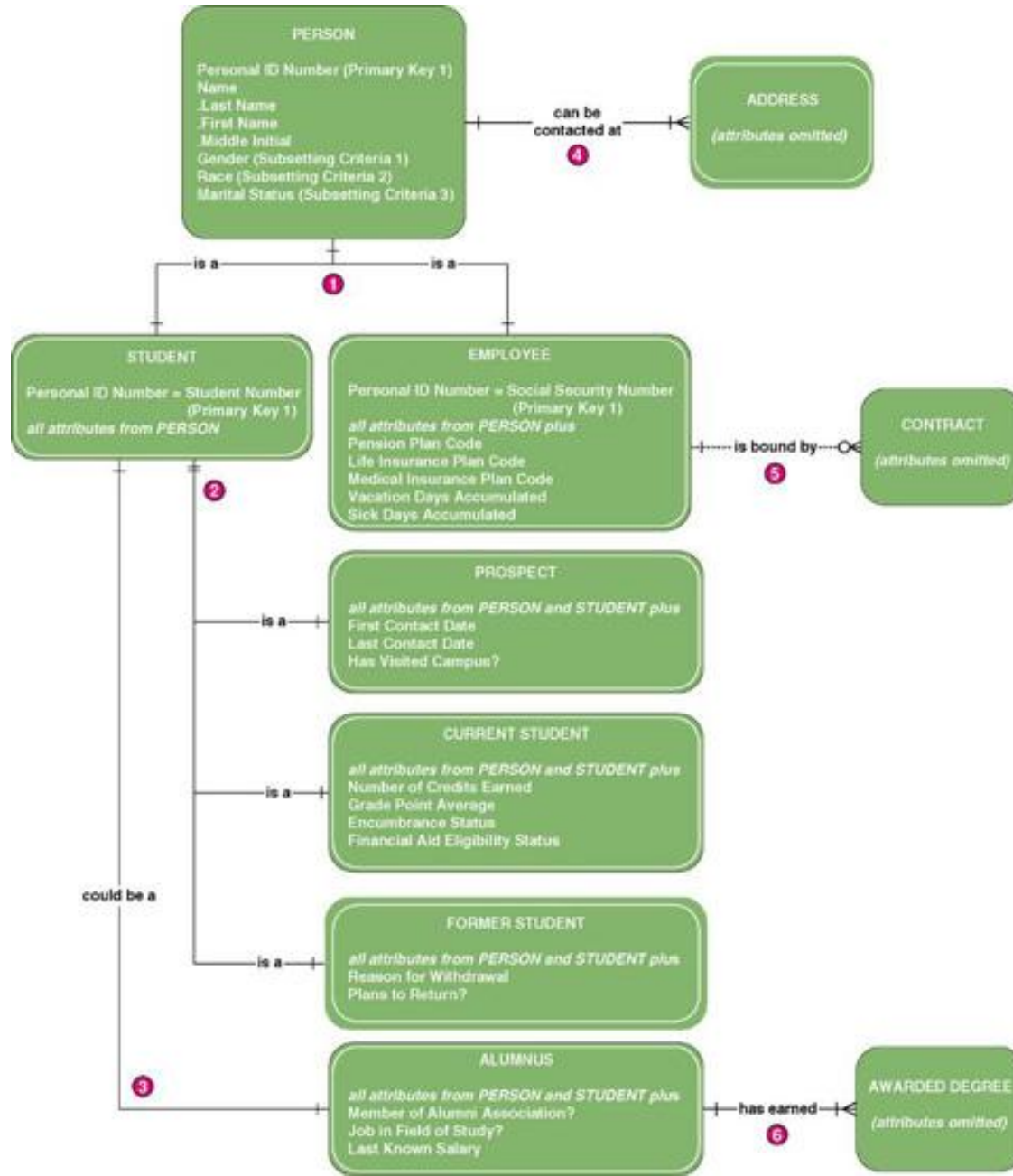
**Subsetting criteria** – an attribute(s) whose finite values **divide** all entity instances into useful subsets.

Sometimes called an **inversion entry**.



# Generalization Hierarchy

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26

# Logical Model Development Stages

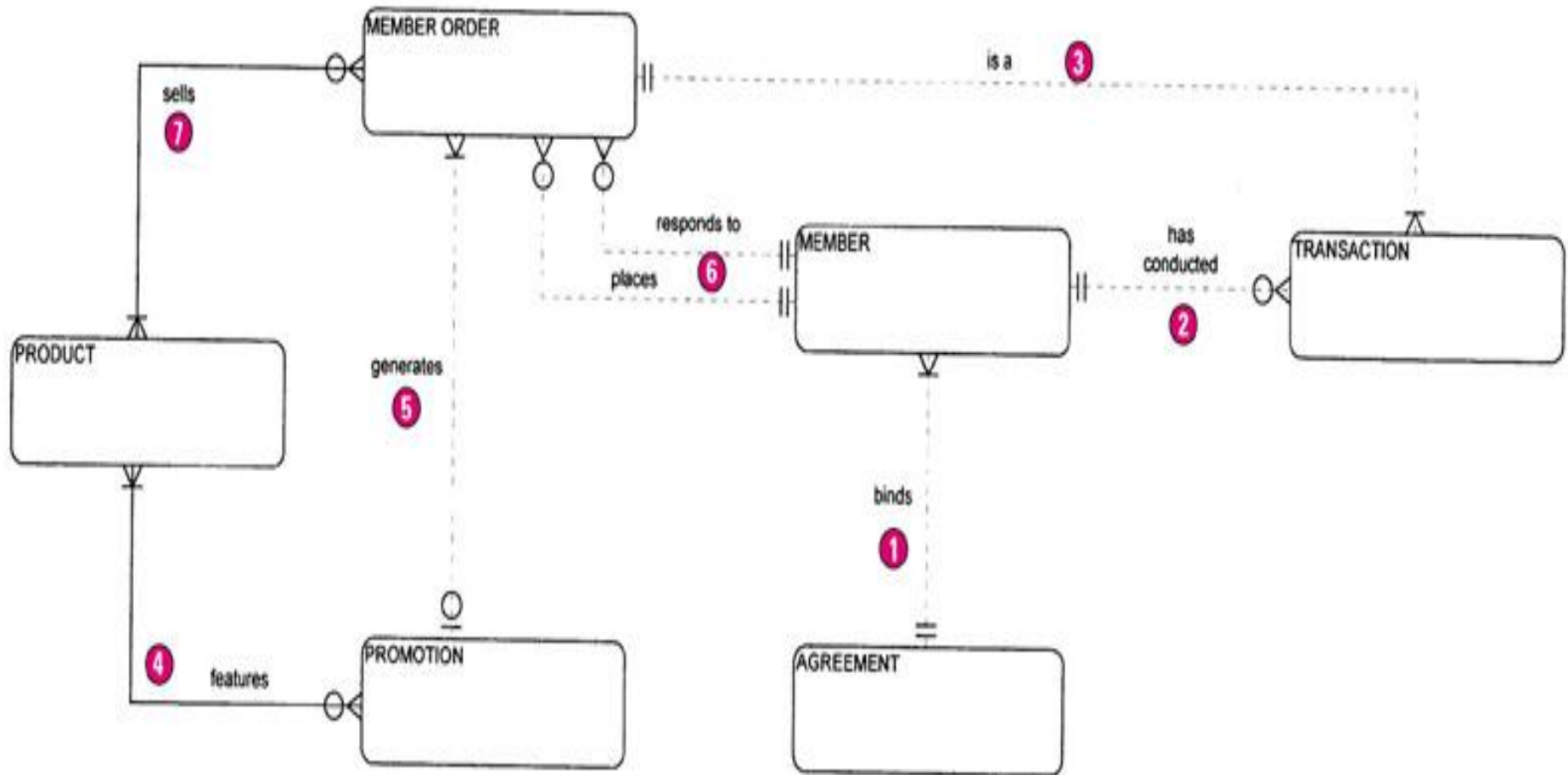
1. **Conceptual data model**
2. **Key-based data model**
3. **Fully attributed data model**

# 1. Conceptual Data model

- Includes only entities and relationships
- To establish project scope

# The Conceptual Data Model

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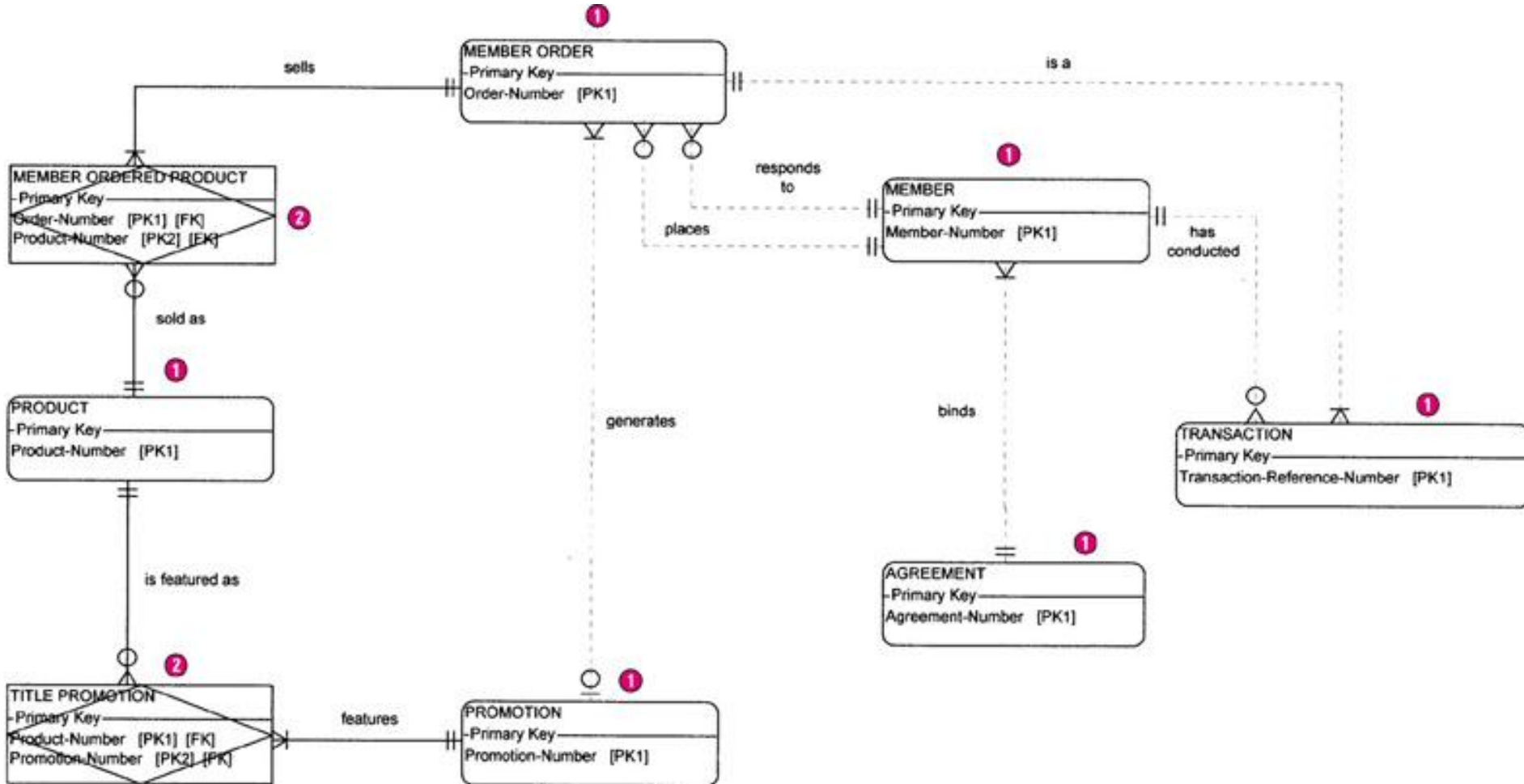


## 2. Key-based data model

- Eliminate nonspecific relationships (many-to-many)
- Add associative entities
- Include primary and alternate keys
- Precise cardinalities

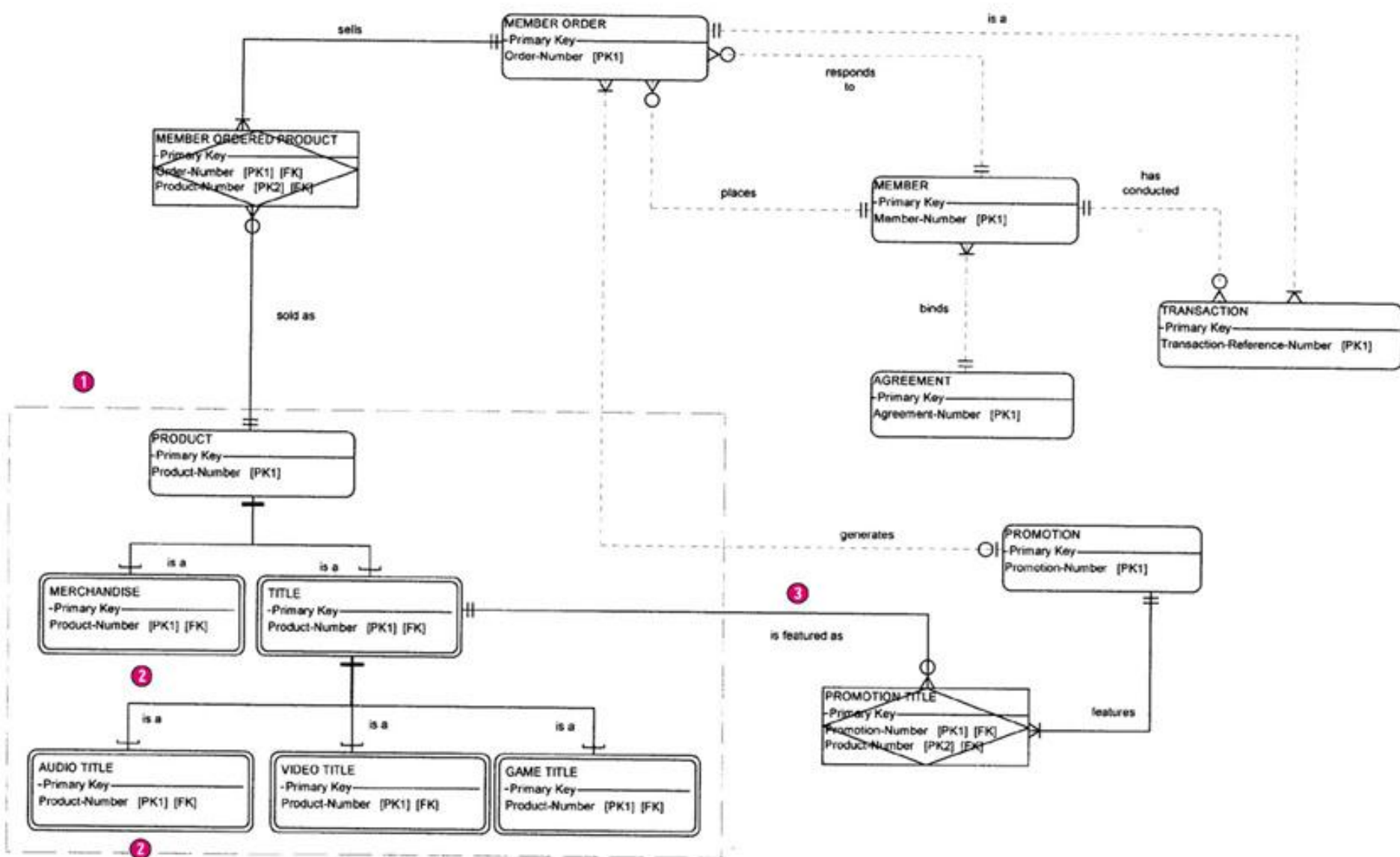
# The Key-based Data Model

5



# The Key-based Data Model with Generalization

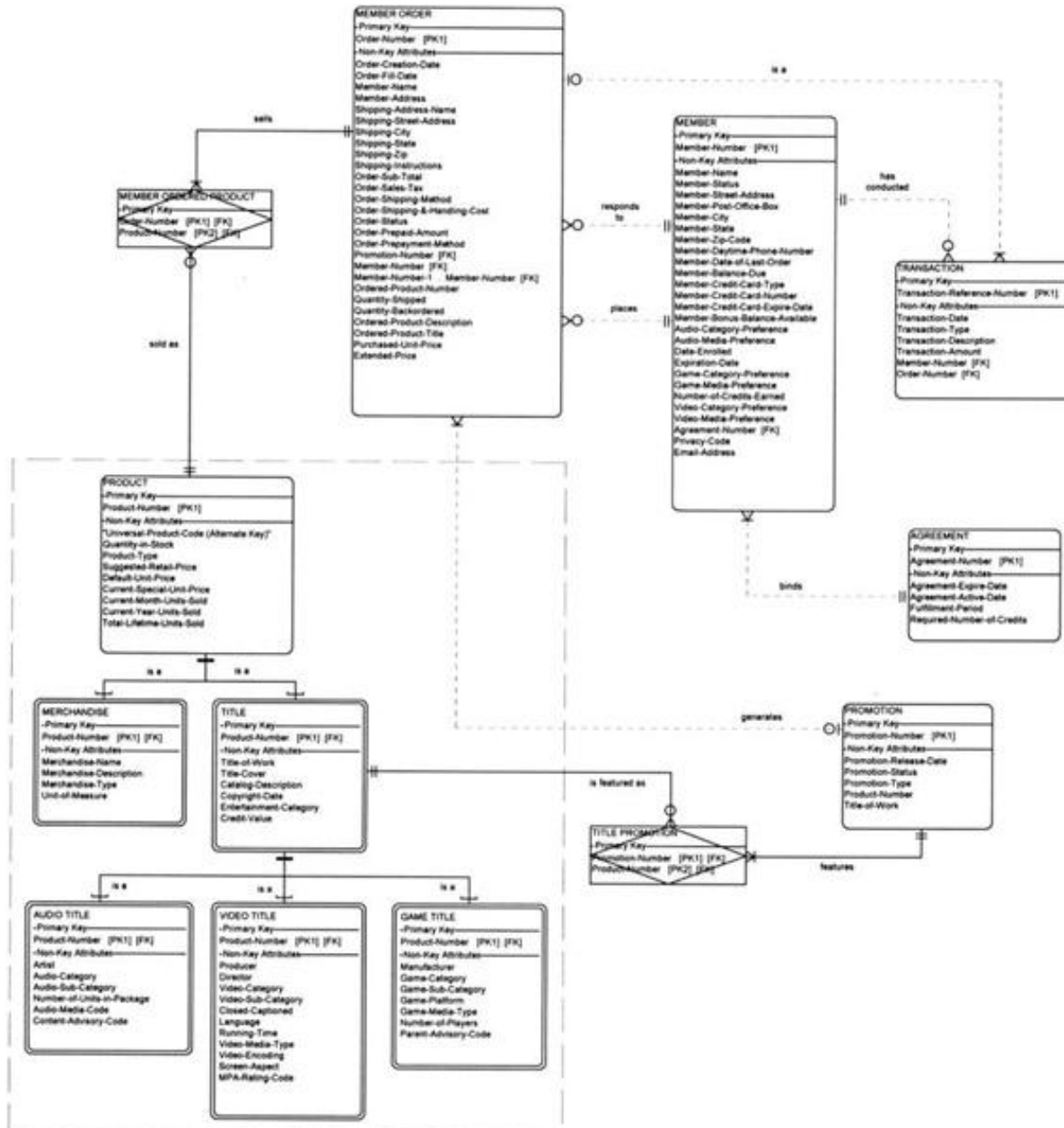
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### 3. Fully attributed data model

- All remaining attributes
- Subsetting criteria

# The Fully-Attributed Data Model



# Interview Questions for Data Modeling

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| Purpose                                    | Candidate Questions                                                                                                                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Discover system <b>entities</b>            | What are the <b>subjects of the business</b> ?                                                                                             |
| Discover entity <b>keys</b>                | What <b>unique characteristic</b> (or characteristics) distinguishes an instance of each subject from other instances of the same subject? |
| Discover entity <b>subsetting criteria</b> | Are there any <b>characteristics of a subject that divide all</b> instances of the subject into useful subsets?                            |
| Discover <b>attributes</b> and domains     | What <b>characteristics describe</b> each subject?                                                                                         |

# Interview Questions for Data Modeling

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| Purpose                                    | Candidate Questions                                                                          |
|--------------------------------------------|----------------------------------------------------------------------------------------------|
| Discover <b>security and control</b> needs | Are there any <b>restrictions on who</b> can see or use the data?                            |
| Discover <b>data timing</b> needs          | How often does the <b>data change</b> ?                                                      |
| Discover <b>generalization</b> hierarchies | Are <b>all instances</b> of each subject <b>the same</b> ?                                   |
| Discover <b>relationships</b>              | What <b>events</b> occur that imply associations between subjects?                           |
| Discover <b>cardinalities</b>              | Is each business activity or event handled the same way, or are there special circumstances? |